Incident Report — QakBot bb02: Trace the Loader DLL to its C2

SOC Analysis & Triage · Easy Operation

1. Header / Metadata

Operation ID	OPS-2026-0423-13
Operation name	QakBot bb02: Trace the Loader DLL to its C2
Started (UTC)	2026-04-23 08:15:00
Closed (UTC)	2026-04-23 11:12:30
Analyst	Ndumiso Mbanjwa
Cohort	May26 Cybersecurity
Difficulty	Easy
Severity	High
Final verdict	True Positive
Status	Closed – Escalated for host isolation and reimage, credential reset for the user, and blocking of the bb02 C2 infrastructure

2. Executive Summary

A purchasing coordinator, dorian.haskell, received a phishing email from invoices@sapplus.net that failed sender authentication and pointed at a password-protected archive. Password protection is the point: it stops the mail gateway and the endpoint scanner from reading the contents, so the file arrives unexamined and the password in the message body turns the user into the unpacking tool.

At 09:03:40 the archive Orig1510220021.zip was opened with 7-Zip from the Downloads folder. Inside was a disk image, which Windows mounts as a drive letter, and at 09:05:00 a shortcut on that mounted volume ran cmd.exe /c E:\qbot.cmd. One minute later the handoff that matters: regsvr32.exe, a signed Microsoft utility that ships with Windows, was run with the silent switch against C:\Users\dorian.haskell\AppData\Local\Temp\tdfasdf.dat. That file is the QakBot loader DLL, given a .dat extension so it does not look like a library, and it never ran as its own process — it executed inside a trusted Microsoft binary, which is what the ATT&CK technique T1218.010 describes.

The network behaviour then identified the bot beyond doubt. Between 11:07:04 and 11:08:19 the process opened TLS sessions to five bare IP addresses, none of them preceded by a DNS lookup, and

all five were refused. At 11:08:47 the sixth attempt, to 185.177.34[.]91:443, succeeded — that is the endpoint the bot settled on. A second success followed at 11:12:30 to 187.198.8[.]241:443. This refuse-then-settle pattern across a hardcoded list is characteristic of QakBot's bb02 wave and is exactly what the platform's “Session-Cycle-To-Bare-Addresses” rule is built to catch. All 6 scored tasks were resolved.

3. Operation Details

Triggering detection	Five rules fired: Registration-Utility-From-User-Directory (High), Removable-Volume-Shortcut-Runs-Shell (High), Session-Cycle-To-Bare-Addresses (High), Egress-Reuses-Open-Session (High) and External-Mail-Auth-Failure-With-Archive (Medium)
Source	invoices@sapplus.net (lure); workstation 10.137.30.42, user dorian.haskell
Destination	185.177.34[.]91:443 (sxeuiqecowj[.]net) — the C2 the bot settled on; 187.198.8[.]241:443 (sageiaeuz[.]us) as a second successful session
User / Host affected	dorian.haskell on 10.137.30.42 (purchasing coordinator)
Artefact	Orig1510220021.zip (password-protected archive) → mounted disk image at E: → E:\qbot.cmd → tdfasdf.dat (QakBot loader DLL) in the user's Temp folder, registered by regsvr32.exe /s
Direction	Inbound phishing email and download; outbound TLS beaconing to five refused addresses then two successful C2 sessions
Tool surfaces used	Security log platform (SIEM), endpoint detection (XDR) and firewall/network telemetry (FW)

4. Investigation & Triage

Task	Objective	What I checked	Finding	Decision & reasoning	Fig. ref
Task 1 – Triage the morning alert	Establish what the alert is actually saying before pivoting	The detection panel: five rules, and the severity spread across 662 logs (9 Critical, 39 High, 55	The rules describe a chain rather than one event: a signed registration utility used from a user directory, a shortcut on a mounted volume	Read together, the rule set already sketches the whole case. Starting from the rule wording rather than from raw logs told me which host, which	Fig. 1

Task	Objective	What I checked	Finding	Decision & reasoning	Fig. ref
		Medium, 8 Low)	spawning a shell, a cycle of sessions to unresolved addresses, and a mail message that failed authentication while pointing at a password-protected archive	user and which three telemetry sources mattered	
Task 2 – Trace the delivery	Identify the sender that started the incident	Mail telemetry for the affected user, filtered to messages that failed sender authentication	invoices@sapplus.net — a delivered external message that failed sender authentication and pointed the recipient at a password-protected archive, matching the External-Mail-Auth-Failure-With-Archive rule	A password-protected archive is an evasion, not a courtesy: the password keeps the gateway and the scanner out while the user is given everything needed to open it	Fig. 1
Task 3 – Catch the execution handoff	Name the binary that executed the payload	The XDR process tree on the workstation, following 7zFM.exe through to the child processes	09:03:40 7zFM.exe opens Orig1510220021.zip from Downloads; 09:05:00 cmd.exe /c E:\qbot.cmd runs from the mounted image; 09:06:00 regsvr32.exe runs with /s against a file in the user's Temp folder	regsvr32.exe is the handoff. It is a legitimate, signed Microsoft utility, so it is trusted by default and application-control policies rarely block it — which is precisely why the loader used it	Fig. 2, 3
Task 4 – Recover the payload	Identify the file that was registered	The full command line on the regsvr32 process record	regsvr32.exe /s C:\Users\dorian.haskell\AppData\Local\Temp\tdfasdf.dat	The payload is tdfasdf.dat. A DLL disguised with a .dat extension in a user-	Fig. 3

Task	Objective	What I checked	Finding	Decision & reasoning	Fig. ref
	and executed		— SHA-256 022cb167a29a32dae848be91aef721c74f1975af151807dafcc5ed832db246b7 is recorded for the regsvr32 binary itself	writable Temp folder is not a normal registration target, and the /s switch suppresses any dialogue the user might have seen	
Task 5 – Pin the C2 beacon	Identify the single endpoint the bot settled on	Network sessions attributed to the regsvr32 process, read in time order	Five TLS attempts to bare addresses were refused between 11:07:04 and 11:08:19 (125.20.84.122, 220.123.29.76, 197.204.233.216, 45.230.169.132, 104.233.202.195). At 11:08:47 185.177.34[.]91:443, resolving as sxeuiqecowj[.]net, returned ConnectionSuccess; a second success followed at 11:12:30 to 187.198.8[.]241:443	The answer is 185.177.34[.]91:443 — the first address that accepted the session. The refused attempts are as diagnostic as the successful one: a hardcoded list being worked through is bot behaviour, not user browsing	Fig. 3
Task 6 – Classify the execution proxy	Map the handoff to ATT&CK	The regsvr32 command line together with the fact that no QakBot process ever appeared in its own right	T1218.010 Signed Binary Proxy Execution: Regsvr32	The technique is defined by the mechanism, not the malware family. Naming it correctly is what lets a detection engineer write one rule that covers every future campaign using the	Fig. 3

Task	Objective	What I checked	Finding	Decision & reasoning	Fig. ref
				same living-off-the-land binary	

5. Threat Intelligence

Indicator (defanged)	Tool	Result / score	Date checked (UTC)	Notes
185.177.34[.]91	VirusTotal	0 vendors flagged it — every engine returns Unrated. AS 57469 Azienda Pubbliservizi Brunico, 185.177.32.0/22, RIPE NCC, IT	2026-09-19	The live C2 the bot settled on has no reputation whatsoever. This is the single most important observation in the threat-intelligence section: the address that mattered most was invisible to reputation data, and only the host's own network behaviour exposed it.
sxeuiqecowj[.]net	VirusTotal	0 of 89 vendors flagged it; VirusTotal applies a “dga” tag	2026-09-19	The name resolving to the C2 is algorithmically generated — the dga tag agrees with what the string looks like. Unflagged, but the pattern itself is the signal: no human registers a name like this.
187.198.8[.]241 (sageiaeuz[.]us)	Internal telemetry	Second successful TLS session at 11:12:30, also on port 443, name equally algorithmic	2026-09-19	Recorded so the containment list is complete. Two live endpoints means blocking only the first would have left the bot connected.
regsvr32.exe (SHA-256 022cb167…db246b7)	VirusTotal	Present on VirusTotal and	2026-09-19	As expected — this is Microsoft's own signed utility. A hash lookup on

Indicator (defanged)	Tool	Result / score	Date checked (UTC)	Notes
		Undetected by every engine		the executing process would have returned a clean verdict and closed the case wrongly; the malicious component is the .dat file it was pointed at.
Five refused addresses	Internal telemetry	125.20.84.122, 220.123.29.76, 197.204.233.216, 45.230.169.132, 104.233.202.195 — all TCP/443, all ConnectionFailed, none preceded by a DNS lookup	2026-09-19	Not indicators of compromise in themselves, but valuable as a behavioural fingerprint: a hardcoded fallback list is what distinguishes a bot from a user's browser.

This is assessed with high confidence as a QakBot bb02 infection delivered by the standard 2026 chain: authentication-failing phishing mail, password-protected archive, disk image, shortcut, and a loader DLL registered through a signed Windows utility. External intelligence contributed almost nothing — the live C2 and its algorithmically generated name are unflagged by every VirusTotal vendor, and the only hash that resolves cleanly is Microsoft's own regsvr32. Treating those clean verdicts as reassurance would have been the wrong call. The verdict rests on behaviour recorded by the organisation itself: a signed utility silently registering a .dat file from a user Temp folder, a shortcut on a freshly mounted image spawning a shell, and one process cycling through six unresolved addresses until one accepted a TLS session.

6. Timeline (UTC)

Time (UTC)	Event
2026-04-23 08:15:00	OUTLOOK.EXE opens on the workstation — the phishing message from invoices@sapplus.net is read
2026-04-23 08:40:00	chrome.exe flagged Suspicious — the archive is downloaded
2026-04-23 09:03:40	High — 7zFM.exe opens C:\Users\dorian.haskell\Downloads\Orig1510220021.zip (T1204.002)
2026-04-23 09:05:00	High — shortcut on the mounted disk image runs cmd.exe /c E:\qbot.cmd (T1204.001)

Time (UTC)	Event
2026-04-23 09:06:00	High — regsvr32.exe /s registers tdfasdf.dat from the user's Temp folder (T1218.010)
2026-04-23 11:07:04–11:08:19	Five TLS sessions to bare addresses refused, no DNS resolution preceding any of them (T1571 behaviour)
2026-04-23 11:08:47	ConnectionSuccess to 185.177.34[.]91:443 (sxeuiqecowj[.]net) — the C2 the bot settled on (T1071.001, T1573)
2026-04-23 11:12:30	ConnectionSuccess to 187.198.8[.]241:443 (sageiaeuz[.]us) — second live session

7. Indicators of Compromise

Type	Indicator (defanged)	Context
Email address	invoices@sapplus.net	Sender of the lure; failed sender authentication, pointed at a password-protected archive — platform Confirmed finding
File name	Orig1510220021.zip	Password-protected archive in the user's Downloads folder containing the disk image
File path	E:\qbot.cmd	Shortcut target on the mounted disk image that launched the command interpreter
File name	tdfasdf.dat	QakBot loader DLL disguised with a .dat extension in the user's Temp folder — platform Confirmed finding
Process	regsvr32.exe	Signed Microsoft utility used to proxy execution of the loader — platform Confirmed finding
IPv4:port	185.177.34[.]91:443	C2 endpoint the bot settled on — platform Confirmed finding; VirusTotal 0 detections
Domain	sxeuiqecowj[.]net	Name resolving to the C2; VirusTotal tags it dga
IPv4:port	187.198.8[.]241:443	Second successful C2 session (sageiaeuz[.]us)
IPv4	125.20.84.122, 220.123.29.76, 197.204.233.216, 45.230.169.132, 104.233.202.195	Fallback list addresses refused on TCP/443, none preceded by DNS

Type	Indicator (defanged)	Context
SHA-256	badf4752413cb0cbdc03fb95820ca167f0cdc63b597ccdb5ef43111180e088b0	cmd.exe process record for the qbot.cmd execution
Host	10.137.30.42 (dorian.haskell)	Infected workstation

STIX 2.1 Bundle


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    "id": "bundle--f52295a9-c7c0-5f64-9b35-7dbba9dbebc5",
    "objects": [
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            "spec_version": "2.1",
            "id": "identity--9dc5ad50-3b6f-5d38-b8d4-9abbfbf2cac5",
            "created": "2026-04-23T09:03:40.000Z",
            "modified": "2026-09-19T16:00:00.000Z",
            "name": "Ndumiso Mbanjwa (May26 Cybersecurity)",
            "description": "SOC analyst incident report \u2014 May26 Cybersecurity cohort,
SOC Simulator operation QakBot bb02: Trace the Loader DLL to its C2.",
            "identity_class": "individual",
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            "created": "2026-04-23T09:03:40.000Z",
            "modified": "2026-09-19T16:00:00.000Z",
            "name": "QakBot",
            "description": "Banking trojan and loader, bb02 distribution wave. Delivered
as tdfasdf.dat and executed via regsvr32.exe on 10.137.30.42, then beaconed over TLS/443
to a hardcoded address list until 185.177.34.91 accepted the session.",
            "malware_types": [
                "trojan",
                "backdoor"
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            "is_family": true,
            "created_by_ref": "identity--9dc5ad50-3b6f-5d38-b8d4-9abbfbf2cac5",
            "object_marking_refs": [
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            "created": "2026-04-23T09:03:40.000Z",
            "modified": "2026-09-19T16:00:00.000Z",
            "name": "regsvr32",
            "description": "Signed Microsoft library-registration utility abused as an
execution proxy: regsvr32.exe /s against a DLL renamed to .dat in the user's Temp
directory.",
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            "modified": "2026-09-19T16:00:00.000Z",
            "name": "System Binary Proxy Execution: Regsvr32",
            "description": "Platform-confirmed classification: regsvr32.exe /s used to
register and execute tdfasdf.dat from the user's Temp folder.",
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                    "source_name": "mitre-attack",
                    "url": "https://attack.mitre.org/techniques/T1218/010/",
                    "external_id": "T1218.010"
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                    "phase_name": "defense-evasion"
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            "created": "2026-04-23T09:03:40.000Z",
            "modified": "2026-09-19T16:00:00.000Z",
            "name": "Phishing: Spearphishing Attachment",
            "description": "Message from invoices@sapplus.net failing sender
authentication and pointing at a password-protected archive.",
            "external_references": [
                {
                    "source_name": "mitre-attack",
                    "url": "https://attack.mitre.org/techniques/T1566/001/",
                    "external_id": "T1566.001"
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                    "phase_name": "initial-access"
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            "created": "2026-04-23T09:03:40.000Z",
            "modified": "2026-09-19T16:00:00.000Z",
            "name": "User Execution: Malicious File",
            "description": "The user opened Orig1510220021.zip with 7-Zip and ran the
shortcut on the mounted disk image.",
            "external_references": [
                {
                    "source_name": "mitre-attack",
                    "url": "https://attack.mitre.org/techniques/T1204/002/",
                    "external_id": "T1204.002"
                }
            ],
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                    "phase_name": "execution"
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            "created": "2026-04-23T09:03:40.000Z",
            "modified": "2026-09-19T16:00:00.000Z",
            "name": "Obfuscated Files or Information",
            "description": "The loader DLL was renamed tdfasdf.dat, and the archive was
password-protected to defeat gateway and endpoint scanning.",
            "external_references": [
                {
                    "source_name": "mitre-attack",
                    "url": "https://attack.mitre.org/techniques/T1027/",
                    "external_id": "T1027"
                }
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            "modified": "2026-09-19T16:00:00.000Z",
            "name": "Application Layer Protocol: Web Protocols",
            "description": "TLS sessions on port 443 to the C2 after the loader
executed.",
            "external_references": [
                {
                    "source_name": "mitre-attack",
                    "url": "https://attack.mitre.org/techniques/T1071/001/",
                    "external_id": "T1071.001"
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            "created": "2026-04-23T09:03:40.000Z",
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            "name": "Encrypted Channel",
            "description": "All beacon traffic used TLS, with two sessions reaching live
endpoints.",
            "external_references": [
                {
                    "source_name": "mitre-attack",
                    "url": "https://attack.mitre.org/techniques/T1573/",
                    "external_id": "T1573"
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            "created": "2026-04-23T09:03:40.000Z",
            "modified": "2026-09-19T16:00:00.000Z",
            "name": "QakBot bb02 C2 endpoint",
```

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            "description": "TLS/443 ConnectionSuccess at 2026-04-23T11:08:47Z after five
refused attempts. VirusTotal 0 detections, AS57469 (2026-09-19).",
            "indicator_types": [
                "malicious-activity"
            ],
            "pattern": "[ipv4-addr:value = '185.177.34.91']",
            "pattern_type": "stix",
            "valid_from": "2026-04-23T09:03:40.000Z",
            "created_by_ref": "identity--9dc5ad50-3b6f-5d38-b8d4-9abbfbf2cac5",
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            "target_ref": "malware--6ac36ce9-f025-516d-90fa-d2471282855a",
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            "created": "2026-04-23T09:03:40.000Z",
            "modified": "2026-09-19T16:00:00.000Z",
            "name": "Secondary C2 session",
            "description": "Second TLS/443 ConnectionSuccess at 2026-04-23T11:12:30Z,
resolving as sageiaeuz.us.",
            "indicator_types": [
                "malicious-activity"
            ],
            "pattern": "[ipv4-addr:value = '187.198.8.241']",
            "pattern_type": "stix",
            "valid_from": "2026-04-23T09:03:40.000Z",
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            "modified": "2026-09-19T16:00:00.000Z",
            "relationship_type": "indicates",
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            ]
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            "modified": "2026-09-19T16:00:00.000Z",
            "name": "Algorithmically generated C2 name",
            "description": "Name associated with 185.177.34.91. VirusTotal 0/89 with a dga
tag (2026-09-19).",
            "indicator_types": [
                "malicious-activity"
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            "created_by_ref": "identity--9dc5ad50-3b6f-5d38-b8d4-9abbfbf2cac5",
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            "modified": "2026-09-19T16:00:00.000Z",
            "name": "QakBot loader DLL",
            "description": "Loader registered from
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23T09:06:00Z.",
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            ],
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            "modified": "2026-09-19T16:00:00.000Z",
            "name": "Delivery archive",
            "description": "Password-protected archive opened with 7-Zip at 2026-04-
23T09:03:40Z, containing the disk image.",
            "indicator_types": [
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            "pattern": "[file:name = 'Orig1510220021.zip']",
            "pattern_type": "stix",
            "valid_from": "2026-04-23T09:03:40.000Z",
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directory.",
            "indicator_types": [
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            ],
            "pattern": "[process:command_line MATCHES 'regsvr32.*\\\\.dat']",
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            ]
        }
    ]
}
```


8. MITRE ATT&CK Mapping

Tactic	Technique	ID
Defense Evasion	System Binary Proxy Execution: Regsvr32	T1218.010
Initial Access	Phishing: Spearphishing Attachment	T1566.001
Execution	User Execution: Malicious File	T1204.002
Defense Evasion	Obfuscated Files or Information	T1027
Command and Control	Application Layer Protocol: Web Protocols	T1071.001
Command and Control	Encrypted Channel	T1573

T1218.010 is the platform's own confirmed answer to the scored classification task. T1566.001, T1204.002, T1027, T1071.001 and T1573 are stated here because each is evidenced directly by a record recovered in the scored tasks — the authentication-failing mail with its password-protected archive, the 7-Zip and shortcut execution, the renamed DLL, and the TLS sessions — rather than inferred from the QakBot family name. This operation's console does not expose an ATT&CK panel of its own.

9. Impact & Containment

Impact: the workstation 10.137.30.42 is fully compromised. QakBot is a loader as much as a banking trojan, so an established C2 session means the operator can deliver whatever comes next — credential theft, reconnaissance, or a hands-on-keyboard intrusion leading to ransomware, which is the usual end of this chain. Everything typed or stored on that host since 09:06 must be treated as

exposed, including the user's credentials and any cached corporate sessions. Two C2 sessions were live, so partial blocking would not have severed control. Containment steps, in priority order:

1. Isolate 10.137.30.42 from the network immediately and preserve memory and disk before any cleanup, since the loader ran in the context of a signed binary and its injected code lives in memory.
2. Block 185.177.34[.]91 and 187.198.8[.]241 on TCP/443 at the perimeter, along with the names sxeuiqecowj[.]net and sageiaeuz[.]us, and hunt the proxy and firewall logs for any other host reaching either address.
3. Reset dorian.haskell's credentials and revoke active sessions from a clean device; treat any credential entered on the host since the infection as compromised.
4. Reimage rather than clean the workstation — QakBot's persistence and follow-on payloads do not justify trusting a repaired host.
5. Search the mail platform for other recipients of invoices@sapplus.net and for any further password-protected archives delivered after an authentication failure, and quarantine them.

10. Conclusion & Recommendations

Final verdict: True Positive. A phishing message that failed sender authentication delivered a password-protected archive containing a disk image; a shortcut on that image ran the command interpreter, which used the signed Microsoft utility regsvr32.exe to register a QakBot loader DLL disguised as tdfasdf.dat; the bot then worked through a hardcoded address list, five refusals in a row, until 185.177.34[.]91:443 accepted a TLS session. All 6 scored tasks were resolved from the platform's SIEM, XDR and network records.

What makes this case worth keeping is how completely reputation data failed. The C2 is unflagged by every vendor, its domain is unflagged, and the executing process was a legitimate Microsoft binary with a clean hash. An analyst working indicator-first would have found nothing to act on. The chain was exposed by structure instead: a signed utility running from a user directory with a silent switch, a shortcut on a freshly mounted image, and a process touching six addresses without a single DNS lookup. Those are properties of how the attack works rather than of who is running it, which is why the detections that caught it will still catch the next wave under different infrastructure.

Recommendations:

1. Block or alert on regsvr32.exe, rundll32.exe and similar signed utilities executing files from user-writable directories such as Downloads, Temp and AppData — legitimate registration happens from Program Files.
2. Prevent automatic mounting of disk images (.iso, .img) downloaded from the internet via Group Policy, which breaks this delivery chain at the container stage.
3. Quarantine password-protected archives from external senders, especially where the message also fails SPF, DKIM or DMARC — both conditions were present here.
4. Alert on any process opening multiple outbound sessions to addresses with no preceding DNS resolution, which is the behavioural signature of a hardcoded C2 list.

5. Treat clean VirusTotal results on infrastructure as inconclusive rather than exculpatory, and make behavioural evidence the basis of the verdict.
6. Hunt historically for the bb02 pattern — regsvr32 against a non-DLL extension — to find infections that never reached a C2 and so produced no network alert.

11. Evidence / Screenshots

Figure	Caption	Referenced in
Figure 1	The five detection rules fired for this incident, with the severity distribution across 662 security logs	Tasks 1, 2
Figure 2	The XDR process tree showing the delivery sequence: Outlook, browser download, 7-Zip opening Orig1510220021.zip, then cmd.exe running E:\qbot.cmd from the mounted image	Task 3
Figure 3	The execution handoff and beaconing: regsvr32.exe /s registering tdfasdf.dat, followed by five refused TLS sessions and the successful connection to 185.177.34.91:443	Tasks 3, 4, 5, 6
Figure 4	The SIEM security-log view (662 entries) with the telemetry sources available for this investigation	Section 4

Figure 1. The five detection rules fired for this incident, with the severity distribution across 662 security logs.

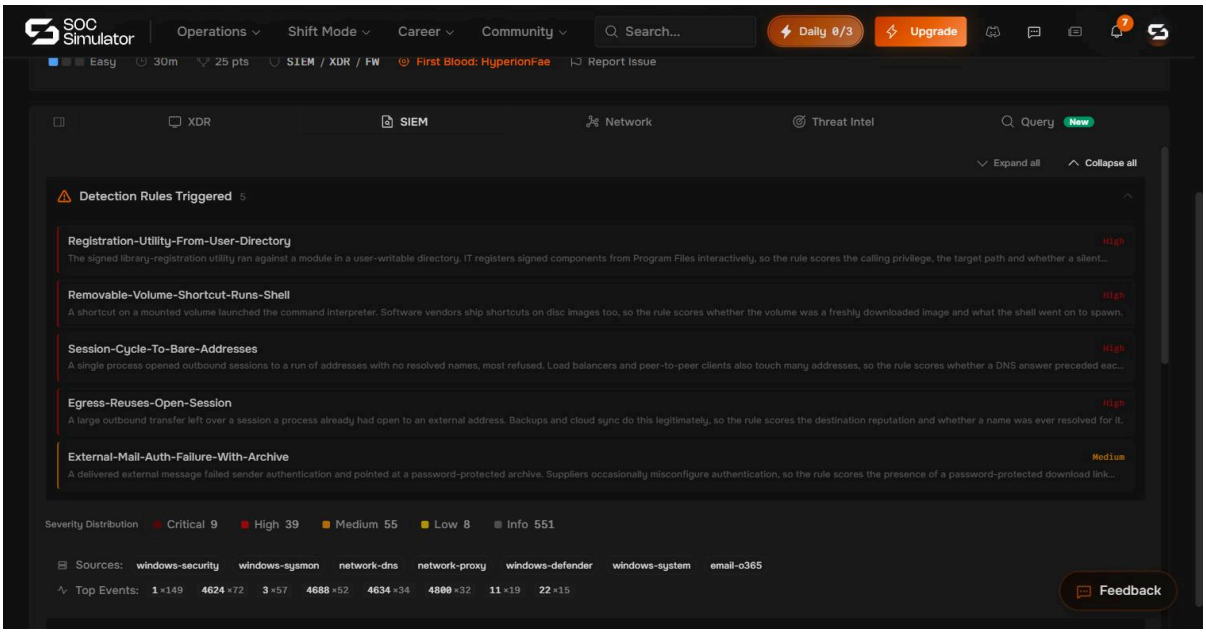


Figure 2. The XDR process tree showing the delivery sequence: Outlook, browser download, 7-Zip opening Orig1510220021.zip, then cmd.exe running E:\qbot.cmd from the mounted image.

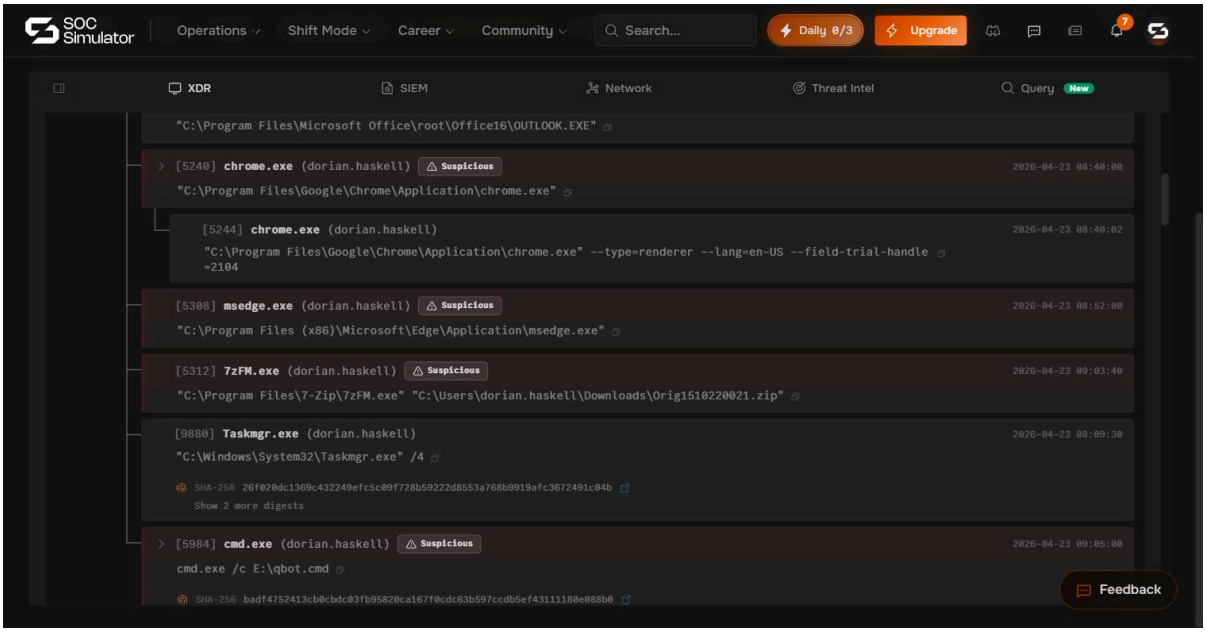


Figure 3. The execution handoff and beaconing: regsvr32.exe /s registering tdfasdf.dat, followed by five refused TLS sessions and the successful connection to 185.177.34.91:443.

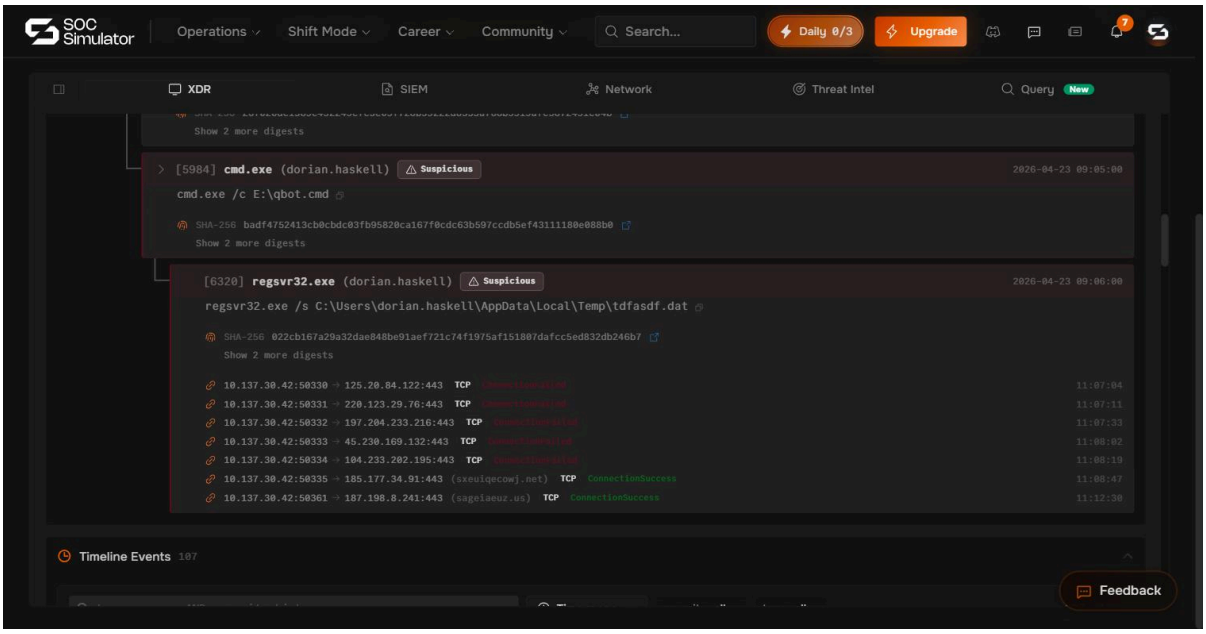


Figure 4. The SIEM security-log view (662 entries) with the telemetry sources available for this investigation.

